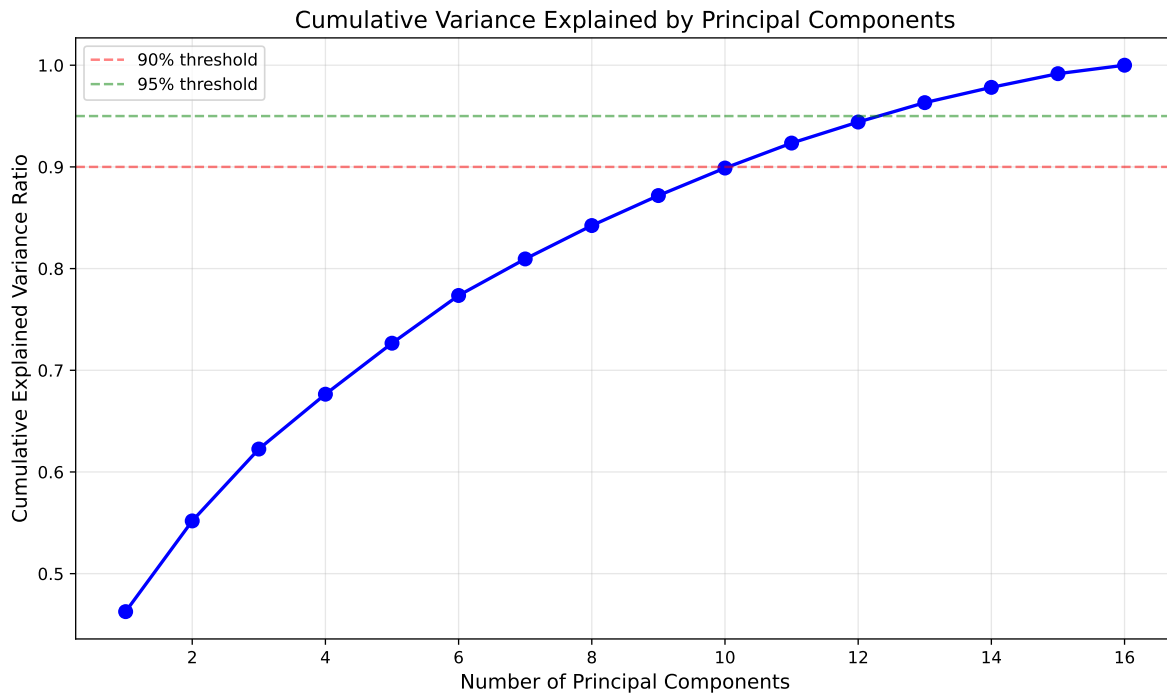
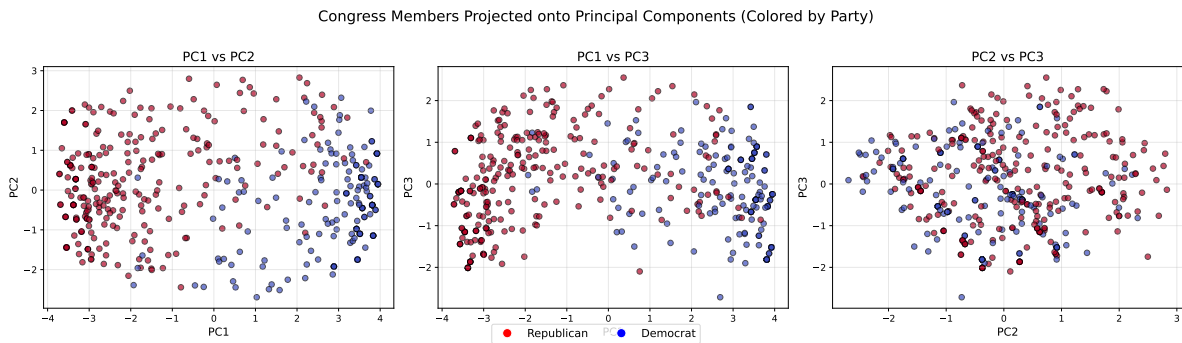


1

a



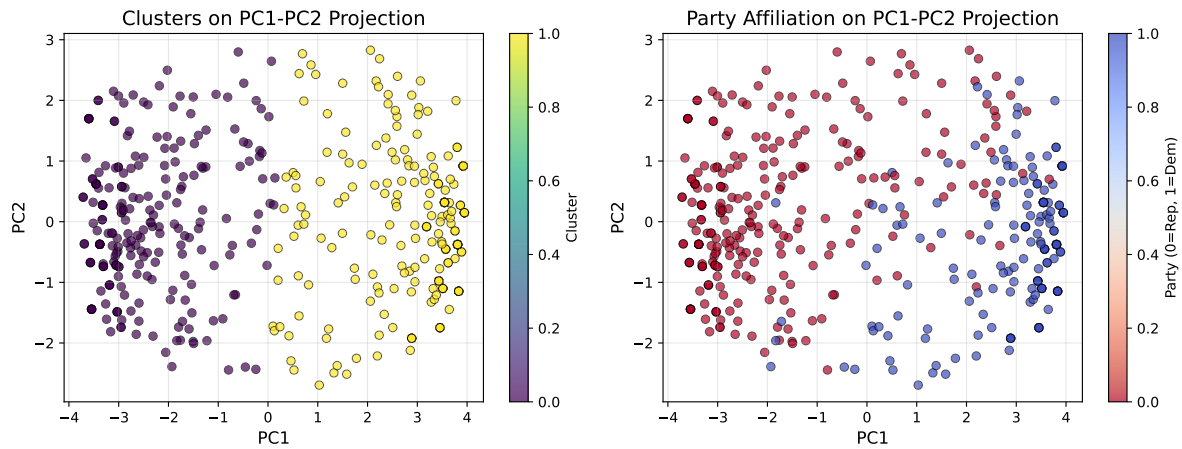
If we are using a threshold of $\alpha = 0.1$ then we need 11 components. If we are instead using $\alpha = 0.05$ we should use 13 components.



The principal components that separate the parties the best is PC1 vs PC2. It appears that the congress members tend to vote similarly if they are with the same party affiliation. We come to this conclusion because our PC's are able to generate a pretty good separation of the two parties.

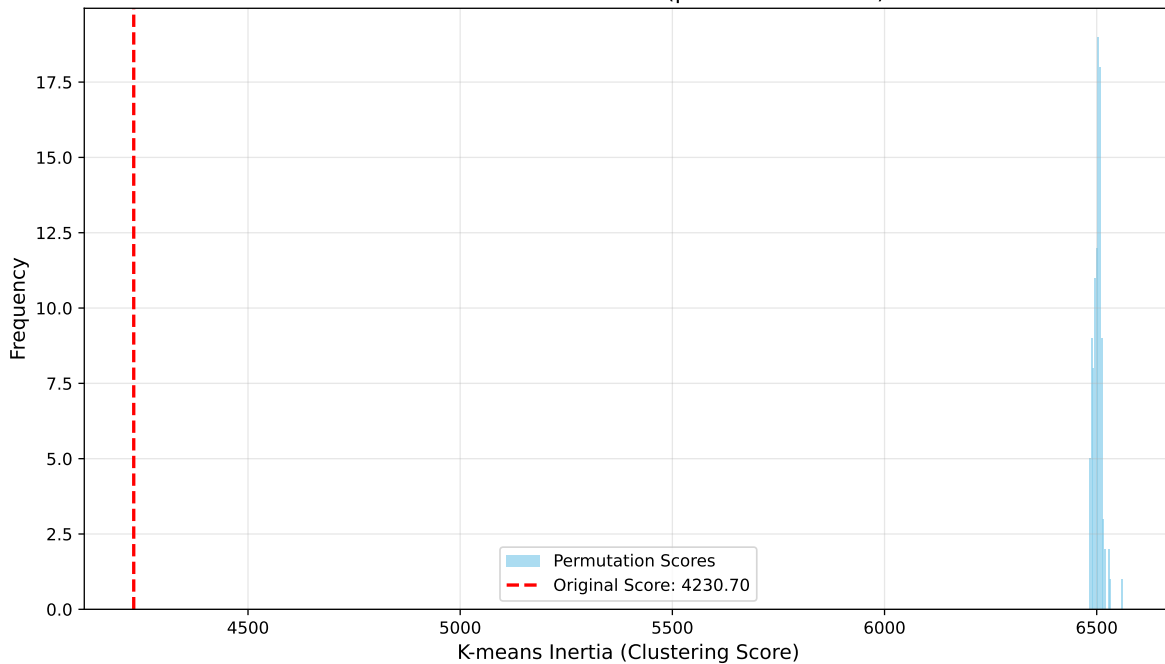
b

K-means Clustering (k=2) vs Party Affiliation



According to the scatterplot, the groups are quite visually separated. Here the cluster groups are roughly split vertically in half where the left cluster group closely alligns with the republican party while the right cluster group closely matches with the democratic affiliation. Note that the clusters are on the 16 voting topics and not on the PC's.

Permutation Test Results (p-value = 0.0000)



The score used to measure the quality of the clustering was the kmeans inertia score. According to the graph, it is quite clear that the clustering is statistically significant and that the congress members votes show meaningful clustering patterns that are unlikely to occur from random chance. With the kmeans inertia score, it measures the squared distance from a point to the center of the cluster, so having a lower kmeans inertia means the clusters are stronger.

c

For quantifying the agreement of clusters and party affiliation, I will use accuracy and mutual information. The clusters had an overall accuracy of 0.8828, meaning they assigned a congress member to the correct group 88.28 of the time. Here the MI was 0.3461.



For quantifying the agreement of the 2 PC clusters and party affiliation, the overall accuracy was 0.8828, meaning they assigned a congress member to the correct group 88.28 of the time and the MI was 0.3461. Additionally, the kmeans inertia was 1118.96, which makes sense as since the dimensions are lower the distances will also decrease.

This means that the original clustering using the 16 votes was more accurate and agreed more with the affiliations than the principal component clustering. This makes sense because the first two PC's only capture roughly 55% of the variation within the data. Since we only have 2PC's we are projecting the 16 dimensional vote space onto 2 dimensions, thus we may be removing some meaningful distinctions.

2

a



The label design is to create a binary response of roughly even split. Here the threshold is quality ≥ 6 , which results in good class balance, interpretability (roughly same amount of quality levels above and below threshold for both wines) and applied evenly to both datasets.

b

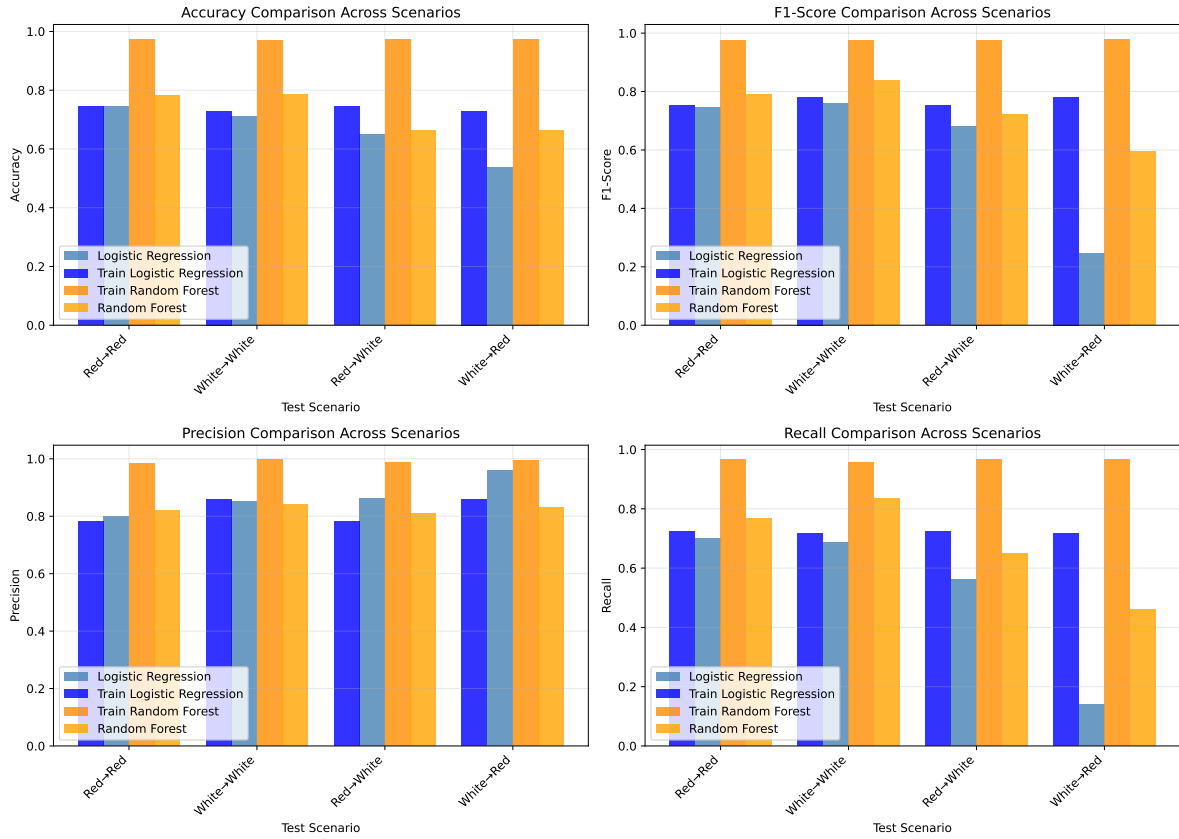
For this section, I used an 80/20 test/training split and the models I chose were Logistic and Random Forest. For the models, I used all of the features and did not perform any sort of feature selection. As mentioned in the problem, since the red wine set has significantly less data points than the white wine set, I decreased the white wine training set to match the size

of the red wine training set (1279 samples). The results of applying the models to the data is shown in the table below.

Testing Scenario	Model	Accuracy	Precision	Recall
Red → Red (In-domain)	Logistic Regression	0.747	0.800	0.702
Red → Red (In-domain)	Random Forest	0.812	0.836	0.807
White → White (In-domain)	Logistic Regression	0.712	0.839	0.702
White → White (In-domain)	Random Forest	0.767	0.836	0.804
Red → White (Cross)	Logistic Regression	0.650	0.864	0.563
Red → White (Cross)	Random Forest	0.699	0.779	0.764
White → Red (Cross)	Logistic Regression	0.559	0.941	0.187
White → Red (Cross)	Random Forest	0.691	0.816	0.544

C

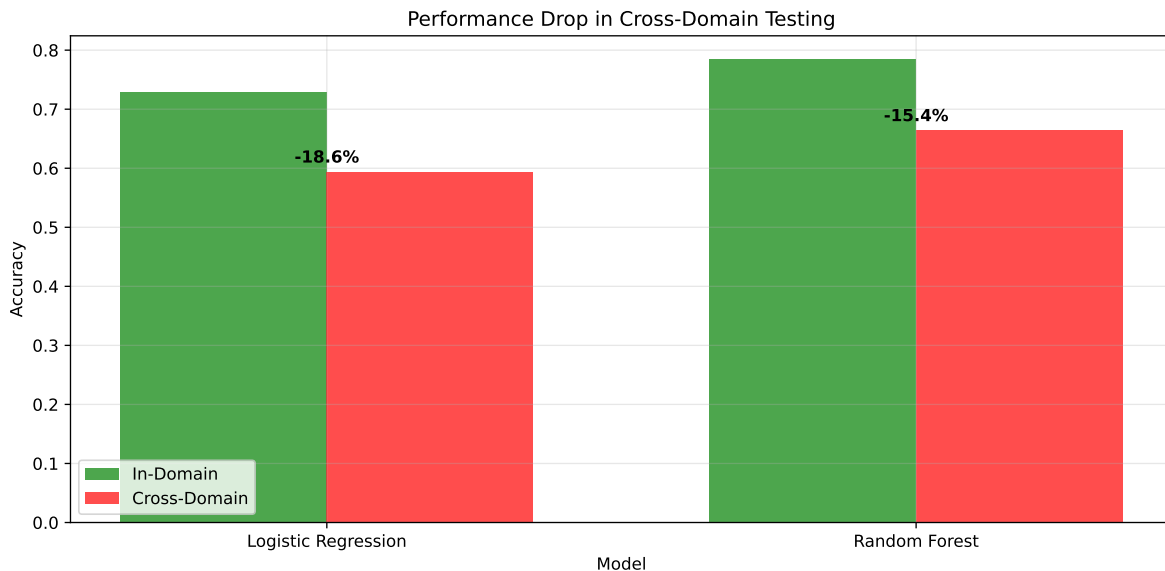
1) In Domain Model Consistency



When looking solely at the in domain scenarios within the graph, we see that Random Forest outperforms the linear regression for all metrics in all in domain scenarios (except for maybe precision in White -> White). However when looking at consistency, it appears the models are similar in consistency with a slight edge to the random forest. If we look at the accuracy graph, the difference between the accuracies in the in domain scenarios for logistic regression is roughly 0.035, while the accuracy difference for the Random Forest is roughly 0.045. For precision, Logistic Regressions had a difference of 0.039 while Random Forest was roughly 0. Lastly, for recall Logistic Regression had a difference of 0 and the Random Forest had a difference 0.03.

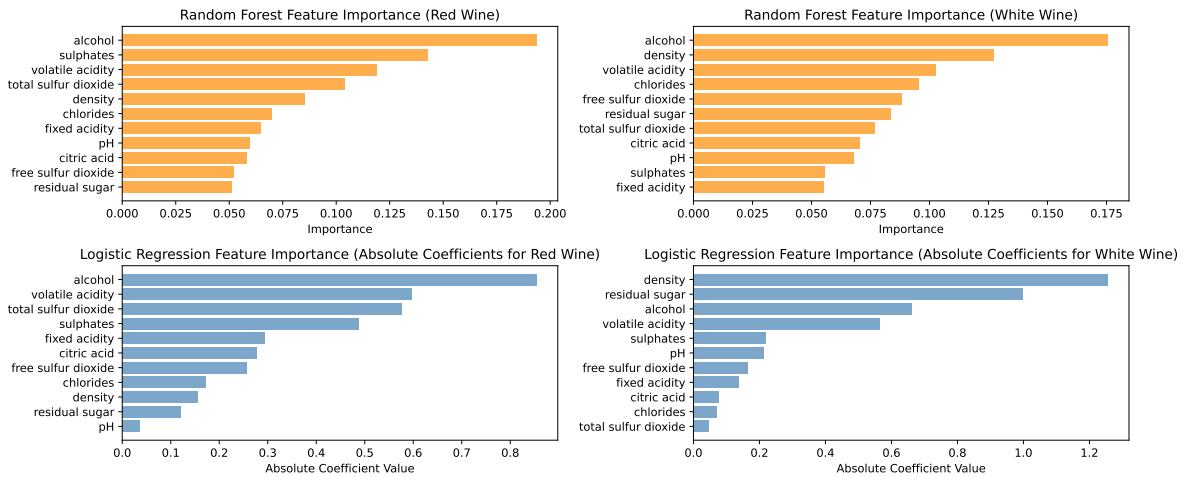
When looking at overfitting, it appears that the logistic regression is not overfitted as the different performance metrics increase and decrease by very small margins between training and testing. Random Forest however, there is significant decrease in all performance metrics when going from training to testing suggesting the presence of overfitting.

2) Cross Domain Model Generalization



Using the information from this graph, the table in 2b, and the graph in 2c1, it appears that the random forest performs slightly better than the logistic regression for cross domain. With the exception of precision, random forest outperforms the logistic regression model in every metric. The logistic regression decreased accuracy by an average of 0.13 whereas the random forest model decreased accuracy by an average of 0.1 when switching from in domain to cross domain.

3) Cross Domain Performance Differences



When trying to determine what factors that could explain the performance differences across domains, perhaps one of the biggest factors is the shift in feature importance across domains. From the graph we can see that both Logistic Regression and Random Forest models both think that the residual sugar is quite important for white wine predictions but not very important for red wine predictions. Essentially, the models do not perform well cross domain because it is weighing factors as important while the factor may not necessarily be important. A good analogy would be determining performance of football athletes based on age, weight, and height. Weight might be an important factor for linemen, but not as important for determining performance of a wide receiver.